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SkySenseGPT：面向遥感视觉-语言理解的细粒度指令调优数据集与模型

Junwei Luo
Wuhan University
luojunwei@whu.edu.cn

Zhen Pang
Wuhan University
pangzhen@whu.edu.cn

Yongjun Zhang
Wuhan University
zhangyj@whu.edu.cn

Tingzhu Wang
Wuhan University
tingzhu.wang@whu.edu.cn

Linlin Wang
Wuhan University
wangll@whu.edu.cn

Bo Dang
Wuhan University
bodang@whu.edu.cn

Jiangwei Lao
Ant Group
wenshuo.ljw@antgroup.com

Jian Wang
Ant Group
bobblair.wj@antgroup.com

Jingdong Chen
Ant Group
jingdongchen.cjd@antgroup.com

Yihua Tan
Huazhong University of Science and Technology
yhtan@hust.edu.cn

Yansheng Li*
Wuhan University
yansheng.li@whu.edu.cn

Abstract

遥感大型多模态模型(RSLMMs)正在迅速发展,并在遥感影像(RSI)理解方面展现出显著能力。然而,由于现有数据集的限制,RSLMMs 在理解复杂遥感场景中物体间丰富语义关系方面存在不足。为释放 RSLMMs 的复杂理解能力,我们提出了一个大规模指令调优数据集 FIT-RS,包含 1,800,851 个指令样本。FIT-RS 涵盖了常见的解译任务,并创新性地引入了从关系推理到图像级场景图生成的多个难度递增的复杂理解任务。基于 FIT-RS,我们构建了 FIT-RSFG 基准。此外,我们建立了一个新的基准,用于评估 LMMs 的细粒度关系理解能力,命名为 FIT-RSRC。基于组合指令数据,我们提出了 SkySenseGPT,其在公开数据集和 FIT-RSFG 上均取得了卓越性能,超越了现有的 RSLMMs。我们希望 FIT-RS 数据集能够增强 RSLMMs 的关系理解能力,并为遥感社区提供大规模细粒度数据源。数据集将在https://github.com/Luo-Z13/SkySenseGPT提供。

1 引言

得益于近期大型语言模型（LLMs）[1, 2] 的快速进步与演化，大型多模态模型（LMMs）[3, 4, 5, 6, 7, 8, 9, 10] 的能力显著提升，取得了显著的进展。基于各种大规模视觉-语言数据集 [11, 12, 13, 14, 15, 16]，现有的通用 LMMs 在多项任务中展现了令人印象深刻的能力，包

*Corresponding author.

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表 1: 与现有遥感指令调优数据集的比较。“从头开始”意味着收集**所有图像**从头开始，而不是结合公开数据集。“细粒度”意味着关系是否是细粒度的，并在数据集和评估中明确使用。

Dataset	From Scratch	Fine-Grained	Total Sample	Tasks
RSICap [23]	×	×	2.6k	Image Caption
H^2RSVLM-SFT [24]	×	×	180k	Image Caption, VQA, Multi-Label Scene Classification, Visual Grounding, other basic tasks
GeoChat [25]	×	×	318k	Image Caption, Region Caption, VQA, Scene Classification, Visual Grounding, Object Detection, Grounding Description, Multi-Task Conversation
MMRS [26]	×	×	1005k	Image Caption, Region Caption, VQA, Scene Classification, Visual Grounding, Object Detection, Multi-Task Conversation
SkyEye-968k [27]	×	×	968k	Image Caption, VQA, Scene Classification, Visual Grounding, UAV Video Caption, Phrase Grounding, Multi-Task Conversation
LHRS-Instruct [28]	×	×	81.8k	Image Caption, VQA, Scene Classification, Visual Grounding, Multi-Task Conversation
FIT-RS (Ours)	✓	✓	1800.8k	图像描述、区域描述、视觉问答、多标签场景分类、目标检测、多任务对话 关系检测、关系推理、对象推理、区域级 SGG、图像级 SGG

括处理复杂推理 [7]、光学字符识别（OCR） [17, 18]、场景图生成（SGG） [19] 等复杂任务。此外，LMMs 已扩展到医疗辅助 [20] 和自动驾驶 [21, 22] 等多个其他领域。

智能遥感图像（RSIs）的解读至关重要且意义深远。它为应用提供了宝贵的信息 [29, 30]。一些研究从开源地理数据中收集了遥感专用的图像-文本数据集 [31, 32, 33, 34]，将基础遥感知识融入模型。此外，现有工作 [23, 25, 26, 27, 28, 24, 35] 提出了包含多种下游解读任务的指令调优数据集，以获得全面的遥感大型多模态模型（RSLMMs）。为了提高模型的解读准确率，引入了多个区域级理解任务 [26, 27]，并且一些数据集设计了简单模板来获取目标之间的基本空间关系 [25]。然而，在面对包含众多目标的复杂场景时，它们还远未实现**细粒度**的理解。“细粒度”指的是准确描述目标之间的精细语义关系，如“在同一车道行驶”，而非简单的基于规则的关系，如“在左侧”。

鉴于上述问题，我们构建了一个大规模细粒度指令调优数据集，命名为 FIT-RS（遥感细粒度指令调优），其中包含 1,800,851 个高质量指令样本，旨在提升 RSLMMs 的细粒度理解能力。如表 1所示，FIT-RS 在多个方面超越了现有数据集。具体而言，FIT-RS 不仅涵盖了高质量的基础理解任务，如平均长度为 619.21 的详细图像描述，还包含了几种新颖的复杂理解任务。基于 FIT-RS 中的任务，我们进一步建立了 FIT-RSFG（遥感细粒度）基准。

此外，我们建立了首个评估 LMMs 遥感关系理解能力的基准，命名为 FIT-RSRC（遥感关系理解）。遵循通用领域流行基准的模式，FIT-RSRC 包含多项选择题，并采用 CircularEval 策略 [36, 19]。此外，FIT-RSRC 基于专家和 GPT-4 创建的常识词表，设计了高质量干扰选项，增加了基准的多样性和公平性。我们的贡献可总结如下：

- 提出了一种大规模指令调优数据集 FIT-RS，包含超过 180 万样本，涵盖基础解析任务及新颖的复杂理解任务。该数据集旨在提升小语言模型（SLM）对复杂场景的理解能力，特别是通过关系推理和场景图生成等任务，增强其对物体关系的感知。
- 我们推出 SkySenseGPT，这是一款全面的 RSLMM，在公开遥感数据集上表现卓越，能够胜任多种复杂理解任务。
- 我们提出了首个用于评估 LMMs 在遥感场景中关系理解能力的基准，命名为 FIT-RSRC。它包含了从常识词表中提取的高质量干扰选项，以及不可回答的问题。我们提出的 SkySenseGPT 在 FIT-RSRC 中达到了 55.5% 的整体准确率，超越了现有的 LMMs 和 RSLMMs。

2 相关工作

2.1 通用大型多模态 Model 的复杂理解任务

现有的 LMM 从自然领域的大规模、高质量多模态数据中获得强大的理解能力。特别是场景图数据的引入 [12, 37, 15] 使通用 LMM 能够进行细粒度任务。例如，Pink [38] 引入了指代理解任务，使模型能够完成视觉关系推理和视觉空间推理等任务。All-Seeing 项目 v2 [19] 提出了一种全图场景图生成任务，并引入 CRPE 基准来评估 LMM 的目标识别和关系理解能力。这些复杂的理解任务涉及对目标之间显式标注关系的建模和评估，从而增强对图像的细粒度理解。

2.2 遥感中的大型多模态模型

现有的 RSLMMs [23, 25, 26, 27, 28, 24] 已经在 RSIs 的对话理解方面做出了贡献，涵盖了许多基本的视觉-语言下游任务。为了增强 RSLMMs 的精确解释能力，H^2 RSVLM [24] 聚焦于不可避免的“幻觉”问题，并提出了 RSSA 数据集以提升 RSLMMs 的自我意识能力。此外，一些工作提出了区域级别的理解任务。例如，Geochat 提出了基于图像的描述生成 [25]，SkyEyeGPT [27] 引入了短语定位任务。然而，现有工作使用简单的模板来创建关系，导致关系不准确，并且无法捕捉与目标的工作或运动状态相关的复杂关系。这种复杂关系对于场景的细粒度理解至关重要。

3 FIT-RS 数据集

考虑到复杂遥感场景中实体间丰富的语义关系，我们旨在通过指令调优增强 RSLMMs 的细粒度理解能力，从而构建 FIT-RS 数据集。FIT-RS 的概览如图 1所示。在接下来的章节中，我们将介绍 FIT-RS 中的数据采集与处理、任务定义与创建，以及 FIT-RSRC 基准的开发。

3.1 数据采集与处理

首先，我们收集了全球范围内大尺寸超高分辨率（VHR）遥感图像（RSIs），并手动为其标注详细的场景图标签，构建了 STAR 数据集 [39]。这些图像涵盖了多种复杂的语义场景（如机场、大坝和港口），其大小从 512×768 到 $27{,}860 \times 31{,}096$ 像素不等。我们的 FIT-RS 数

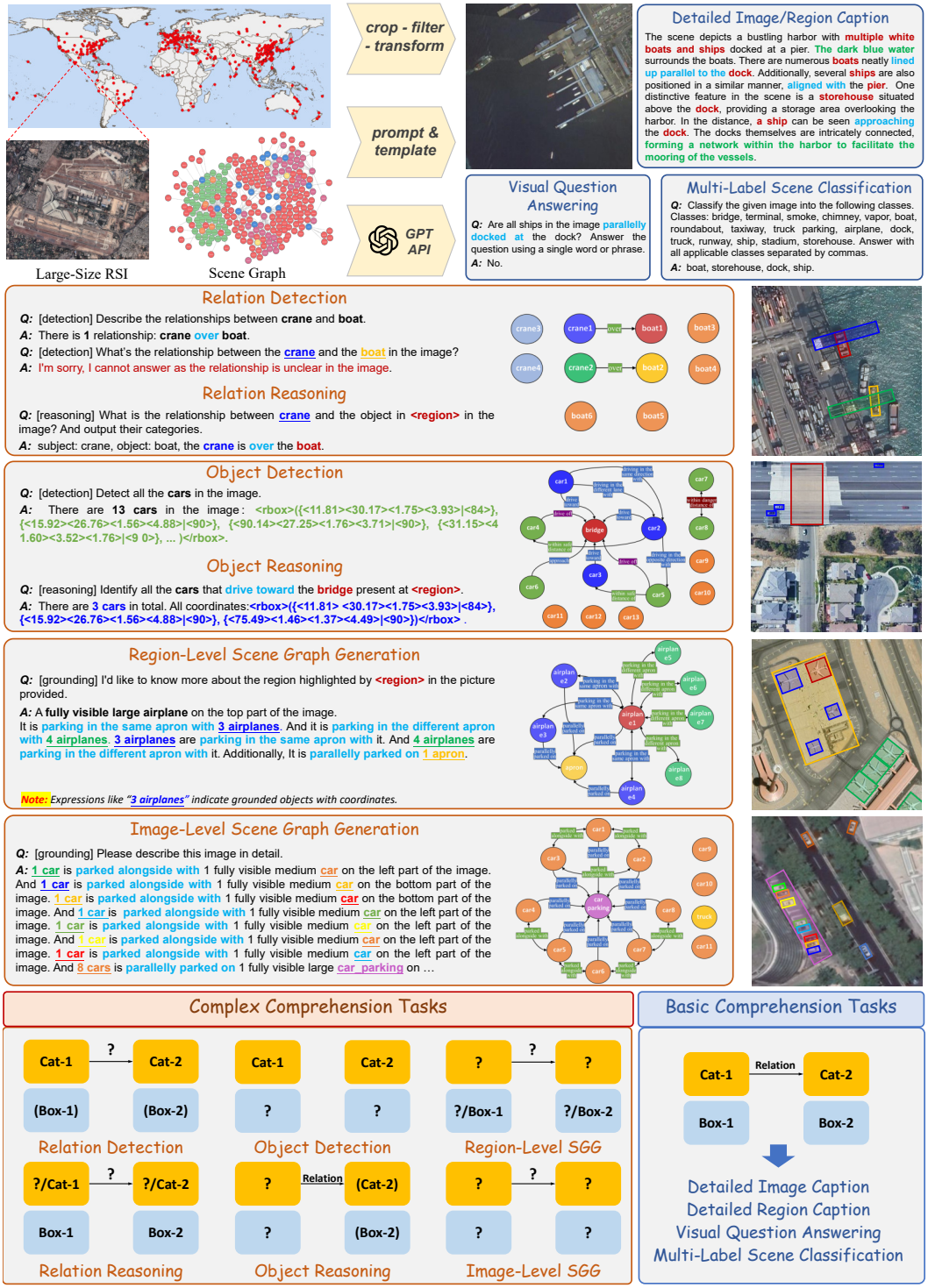

图 1: FIT-RS 数据集概览。图的右上部分和中间部分展示了每个任务的示例。**加粗下划线**表示目标的类别和边界框，即目标定位。下方部分展示了每个任务的设计方法，基本理解任务基于原始场景图标签，而复杂理解任务则构建为一系列逐渐复杂的任务。"?" 表示需要由模型输出的内容，"()" 表示将在输入中随机提供的内容。

据集是 STAR 的扩展，包含了 1,273 幅 RSIs 中的超过 400,000 个三元组和 210,000 多个带有旋转边界框（RBoxes）的对象。它包括 48 类重要目标和 58 类高价值语义关系。具体而言，语义关系被定义为 8 大类：“距离预警”、“空间拓扑”、“功能描述”、“电路布局”、“运动状态”、“排放状态”、“建设状态”和“停放状态”，共包含 58 个子类别。这些关系深入刻画了上下文中的目标。在场景中，多样化的目标共享丰富的类间和类内关系，为细粒度场景理解提供了基础支持。由于现有图像-文本视觉编码器 [40] 的分辨率有限，我们采用常用的裁剪策略 [41] 来处理大尺寸图像。使用 512 × 512 像素的滑动窗口，并设置 100 像素的重叠，将原始大尺寸 RSIs 裁剪为小块。然后，我们通过完整性阈值筛选目标和关系，最终获得 82,532 个有效的图像块。

如图 1 所示，在场景图中，当考虑物体位置时，三元组可以表示为：<subject-box, relation, object-box>。这些元素构成了理解物体间交互关系的基础。然而，RSLMMs 难以直接理解整个场景图。为了更好地探索模型对细粒度理解的能力，我们基于上述基本元素定制了一系列指令跟随任务。对于**基础理解任务**，我们从所有场景图信息中构建指令。对于**复杂理解任务**，我们调整输入问题中的已知元素，并要求模型提供未知元素。这种方法能够创建一组逐步复杂的任务，从而促进 LMMs 对复杂遥感场景的细粒度理解。表 2 展示了 FIT-RS 数据集中样本的详细信息。

表 2: FIT-RS 中指令微调样本的详细信息。缩写及其全称如下：IC：图像描述，RC：区域描述，VQA：视觉问答系统，SC：场景分类，CC：复杂理解，MT：多轮对话。

Task Type	Task	Sample Number
IC	Detailed Image Caption	82532
RC	Detailed Region Caption	90744
VQA	Visual Question Answering	498565
SC	Multi-Label Scene Classification	165074
CC	Relation Detection	181799
	Relation Reasoning	124496
	Object Detection	140626
	Object Reasoning	295395
	Region-Level Scene Graph Generation	93199
	Image-Level Scene Graph Generation	62945
MT	Multi-Turn/Task Conversation	65476

3.2 基础理解任务的数据构建

类似于常见的遥感视觉 - 语言任务，我们构建了几个基本的理解任务。然而，我们的不同之处在于通过充分利用细粒度关系来提高指令的质量和丰富度。

- **详细图像与区域描述**。得益于场景图在对象间高质量详细标注关系，我们能够获得图像的深入描绘。**区域描述**任务紧随其后的区域级 SGG，但移除了边界框。对于详细的**图像描述**任务，生成流水线如下。

  (i) 首先，我们从裁剪后的图像块中提取有效三元组作为前景信息。(ii) 然后，我们使用高效的 TinyLLaVA-3.1B [42] 快速为大量遥感图像生成简洁的背景描述。(iii) 最后，我

们将它们结合作为提示，利用 GPT-4/GPT-3.5 [1] 获得详细且流畅的句子。通过这种方式，我们总共获得了 82,532 个图像-描述对，其中 19,874 个由 GPT-4 生成，62,658 个由 GPT-3.5 生成，GPT-4 生成的描述平均字符长度为 892.66，整体平均长度为 619.21，显著超越了现有的遥感图像-文本数据集（例如，RS5M [31]: 87，SkyScripts [32]: 56，HqDC-1.4M [24]: 369）。

- **视觉问答系统**。得益于物体之间丰富的关联关系，我们能够进行更为精细的视觉问答。相较于传统的视觉问答任务，我们还引入了基于关系的问答系统。

  具体而言，我们设计了以下四类问题。(i) **存在性问题**。这类问题针对特定类别的存在性进行询问。我们平衡正负回答，以避免过多的否定答案。(ii) **比较问题**。我们遵循现有数据集 [43]，比较两个类别的数量。(iii) **计数问题**。我们设计单类别的计数问题，并引入关系约束，询问图像中特定三元组的数量。(iv) **关系问题**。我们设计针对关系多样性的问题。我们选择一个主体-客体类别对，询问特定关系是否存在，或选择一个三元组，询问该关系的唯一性是否满足。总之，FIT-RS 中的 VQA 任务相比现有数据集更为细致，更全面地测试了 RSLMMs 对图像的细粒度理解能力。

- **多标签场景分类**。我们首先手动将原始遥感图像（RSIs）划分为 10 个主要场景类别。随后，设计基于数量的过滤规则，从图像块中选择主要目标类别，这些类别在多标签情境下作为次级类标记。最终的场景分类任务包含两种指令类型：单标签场景分类和多标签目标分类。

3.3 复杂理解任务的数据构建

细粒度的场景理解能力对于 RSLMMs 至关重要，而现有数据集仅提供了有限的理解任务，例如视觉定位和区域描述。我们进一步探索了除上述基本任务之外更具挑战性的任务。具体而言，为了充分利用场景图中的三元组 <subject, relation, object>，我们建立了一系列复杂的理解任务，依次聚焦于关系理解、主体/客体理解以及整体三元组解释，如图 1 下半部分所示。接下来，我们将详细介绍这些复杂的理解任务。

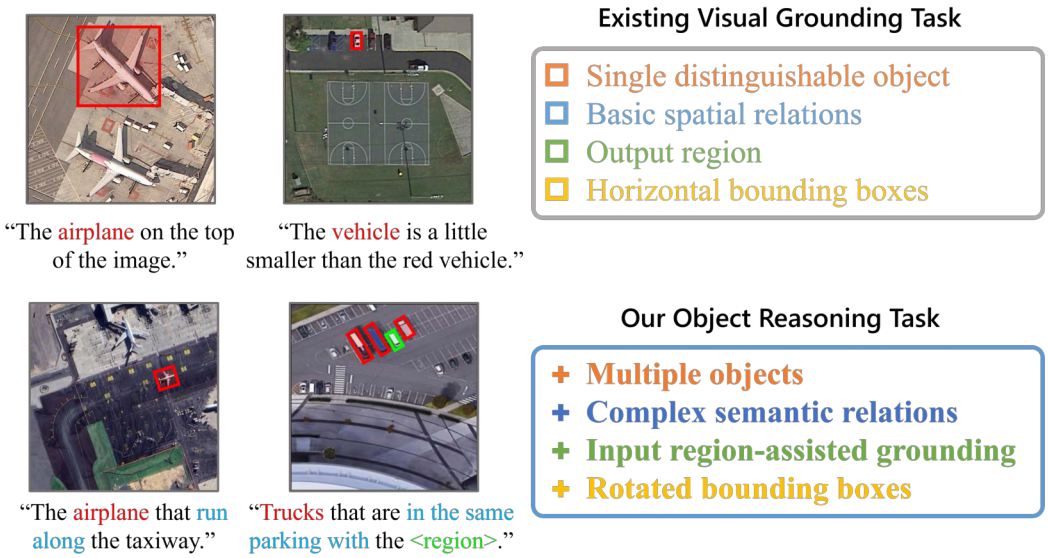

图 2: 传统遥感视觉定位任务与我们提出的细粒度目标推理任务之间的对比。

- **关系检测与推理**。得益于场景图中精确标注的丰富语义关系，我们设计了两项任务以增强 RSLMM 对目标间视觉关系的感知能力。从主体和客体的类别与位置信息中，我们随机选取部分信息作为不完整输入。关系检测任务要求模型仅输出实体间的关系。关系推理任务则进一步要求模型输出主体和客体的类别。关系由 <rel></rel> 标记。
- **目标检测与推理**。这两项任务涉及基于区域的理解和输出。考虑到 RBoxes [44, 45, 46] 的准确率，我们采用“oc”定义来表示精确的 RBoxes，并使用特殊标签 <rbox> 来标记实体的位置，格式为：<rbox>(<cx><cy><w><h>|<angle>)</rbox>。其中，坐标被规范化为 [0.00, 100.00]，保留两位小数，角度以整数度表示。(i) **目标检测**旨在让模型检测并输出特定类别的所有坐标。(ii) **目标推理**更具挑战性：给定特定关系及随机的类别和位置作为查询，模型需要输出满足条件的所有目标的类别和位置。该任务支持区域输入和关系查询，例如“提供图像中位于 <region> 区域且 <rel> 平行停放在 </rel> 停机坪上的所有飞机的坐标。”，帮助用户实现更精确的目标解释。如图 2所示，与传统的遥感视觉定位任务 [47] 相比，我们的目标推理任务具有某些优势。
- **区域级与图像级场景图生成**。这两项任务相对最具挑战性，旨在探索 RSLMMs 在细粒度理解能力上的上限。(i) **区域级** SGG 要求模型根据输入区域描述目标的相对大小、位置及可见性，随后生成所有相关三元组，使用“它”来指代目标；(ii) **图像级** SGG 则涉及直接为整幅图像生成场景图，对于无关联的孤立目标，以目标检测的形式附加到答案中。已定位的目标表示为 <ref> 目标 ></ref><rbox> 框 </rbox>。

  针对上述所有任务，我们构建了相应的负例，即在涉及背景图像时指示拒绝回答的指令，以增强模型的稳定性。所有复杂理解任务的模板及负例响应均可在附录中查阅。此外，我们随机混合不同任务，生成了多轮对话数据。

3.4 遥感关系理解评估

鉴于目前缺乏公开可用的基准来全面、定量地评估现有 LMM 在遥感关系理解中的表现，我们提出了 FIT-RSRC（遥感关系理解）基准。该基准以单选题的形式设计，包含四种不同类型的问题以及高质量干扰选项，如图3所示。遵循主流通用基准的做法，FIT-RSRC 采用 CircularEval [36] 作为评估策略，以增强公平性。

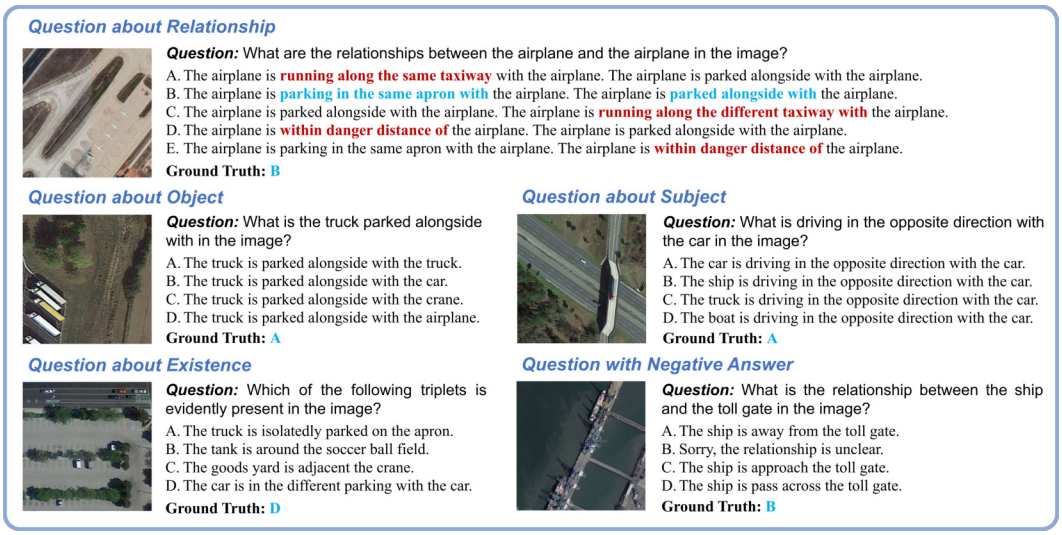

图 3: 我们提出的 FIT-RSRC 中包含若干实例。FIT-RSRC 囊括了四种问题类型，并采用 CircularEval 评估策略，配备了高质量干扰项（源自专家常识词汇集）及不可答问题。

为确保干扰项的高质量，所有错误选项均在预设的常识词汇表范围内生成，该词汇表由 GPT-4 结合专家人工筛选创建。这确保了仅凭常识表达即可判断的错误选项，如“滑行道沿着仓库延伸”，不会出现。如图3中的第一个示例所示，我们基准中的每个样本包含一个正确选项和三个干扰项。若正确答案涉及多重关系，错误答案的长度将与之保持一致。同时，错误选项可能融入正确内容以提升干扰项质量。此外，为评估模型真实性及其对负面图像的鲁棒性，我们引入了不可回答的答案，从而将 LMMs 在负面实例上的幻觉能力纳入定量评估之中。

4 SkySenseGPT 模型

SkySenseGPT 架构遵循主流 LMMs，由视觉编码器（即分辨率为 336 × 336 的 CLIP-ViT-L14 [40]）、作为多模态投影器的多层感知机（MLP）以及 LLM（Vicuna-v1.5 [48]）组成。输入图像按照 Geochat [25] 调整为 504 × 504 像素。我们遵循 LLaVA-v1.5 [7] 获取预训练的 MLP 投影器。在指令调优阶段，我们冻结视觉编码器，微调投影器，并采用 LoRA [49] 来微调 LLM。

我们将 FIT-RS 数据集按 6:2:2 的比例划分为训练集、验证集和测试集。我们使用训练-验证集进行指令调优，并考虑到 LMM 推理的效率，从测试集中随机抽取一部分进行评估。通过这种方式，我们获得了 1415k 个指令样本用于训练。此外，我们还额外收集了现有的公开数据集，并将其转换为最终的指令调优数据用于训练。具体来说，我们收集了 3 个场景分类数据集（NWPU [50]、UCM [51] 和 RSITMD [52]）、3 个 VQA 数据集（EarthVQA [53]、Floodnet-VQA[54] 和 RSVQA-LR [43]）以及 3 个目标检测数据集（DOTA-v2.0 [41]、DIOR [55] 和 FAIR1M [56]）来丰富指令数据集。这部分额外数据包含 365k 个样本。详细信息可在附录中找到。

5 实验

在本节中，我们基于 FIT-RS 数据集和 FIT-RSRC 基准展示了 FIT-RSFG 基准的实验结果。此外，我们使用 SkySenseGPT 对现有公共数据集进行了泛化实验。所有实验均在 4 块 NVIDIA A100 GPU（40G）上进行。所有测试集的具体细节可在附录中找到。

5.1 基准

我们采用现有主流模型对 FIT-RS 数据集中的所有任务进行评估，构建了 FIT-RSFG 基准。为建立 FIT-RSRC 基准，我们广泛评估了多种 LMM 及 RSLMM，验证了 FIT-RSRC 的可靠性与易推广性。

5.1.1 FIT-RSFG 基准

为了构建 FIT-RSFG 基准，我们采用 LLaVA-v1.5 [7] 和 Geochat [25] 作为基准模型，并评估 SkySenseGPT 的微调结果。对于目标定位相关任务，我们采用常见的 mAP [30] 指标，IoU 阈值为 0.25。对于 SGG 任务，我们采用召回率 [57, 58] 和平均召回率指标 [59, 60]。我们使用 GPT-3.5 获取 GPT 评估得分，其值域为 0 到 10。结果如表 3所示。经过微调后，模型能够处理复杂任务，相比基准方法显示出明显改进。详细结果可参见附录。

5.1.2 FIT-RSRC 基准

由于单选格式的简洁性，我们评估了几种通用领域 LMM，包括 LLaVA-1.5-7B [7]、LLaVA-HR-7B [61] 和我们在构建图像描述时使用的 TinyLLaVA [42]。我们还在 FIT-RSRC 上评估了目前开源的 RSLMM，如 GeoChat [25]。

表 3: FIT-RS 中各项任务测试集上基准模型的结果。报告了 zero-shot 和微调的结果。“ZS” 代表 zero-shot 情景，“FT” 代表微调情景。“Failed” 表示由于指令格式差异或模型能力不足，结果无法正确评估。

Task Type	Task	Metric	LLaVA1.5-7B (ZS)	GeoChat (ZS)	SkySenseGPT (FT)
IC	Detailed Image Caption	BLEU-1	15.38	8.79	27.31
RC	Detailed Region Caption	BLEU-1	14.70	14.55	75.82
VQA	VQA	Ave Acc	58.59	53.47	79.76
SC	Multi-Label Scene Classification	Scene Acc	56.18	42.27	82.23
		Obj F1-Score	12.84	14.95	71.43
CC	Relation Detection	F1-Score	Failed	Failed	88.68
	Relation Reasoning	F1-Score	Failed	Failed	74.33
	Object Detection	mAP	Failed	6.13	27.40
	Object Reasoning	mAP	Failed	Failed	5.71
	Region-Level SGG	Recall	Failed	Failed	17.01
	Image-Level SGG	Recall	Failed	Failed	9.60
MT	Multi-Turn/Task Conv	GPT-eval	2.74	3.64	6.60

表 4: 采用 CircularEval 策略的 LMMs 在 FIT-RSRC 上的准确率。“TL” 代表跨任务迁移学习，表示该模型已接触过来自同一源训练集的图像，但此前未遇到过此类特定任务。

Method	Setting	Subject	Object	Relation	Existence	Ave. Acc
LLaVA-1.5-7B [7]	ZS	6.4	11.2	21.0	7.8	11.6
TinyLLaVA [42]		22.0	18.0	12.0	55.6	26.9
LLaVA-HR-7B [61]		7.8	13.4	32.6	47.8	25.4
GeoChat [25]		7.6	8.6	25.6	45.6	21.9
SkySenseGPT	TL	25.2	54.0	50.4	92.2	55.5

结果如表 4所示。基于这些结果，现有的 LMMs 普遍展示出识别三元组存在与否的良好能力，表明其在遥感场景中识别对象的基本能力。然而，当涉及到与对象关系相关的细粒度理解时，现有模型的性能普遍较差。值得注意的是，LLaVA-HR 在 zero-shot 情景下实现了高准确率，表明高分辨率输入对于 RSIs 的细粒度理解具有重要意义。尽管 SkySenseGPT 是为跨任务迁移学习而设置的，但其在识别关系方面表现出优于其他模型的能力。

5.2 泛化实验

我们将 SkySenseGPT 与最先进的 RSLMMs 在公开数据集上进行比较，如表 5所示。在场景分类任务中，我们选择了四个具有代表性的数据集 SIRI-WHU [62]、AID [63]、WHU-RS19 [64] 和 AID-Multilabel [65] 进行 zero-shot 泛化实验。在视觉问答（VQA）任务中，我们选择了两个代表性数据集：RSVQA-HR 和 RSVQA-LR [43]。在场景分类任务中，SkySenseGPT 超越了当前的 RSLMMs，尤其是在 SIRI-WHU 数据集上，超过了最先进的 H^2 RSVLM [24]，准确率提高了 6.25%。在 VQA 任务中，我们的模型在低分辨率和高分辨率

表 5: 在公开数据集上对 RSLMMs 进行比较。对于 VQA 数据集，与 Prior 工作 [23, 25, 24] 类似，我们在评估过程中省略了面积和计数问题。

Task	Setting	Dataset	Metric	Geo-Chat [25]	LHRS-Bot [28]	H^2RS VLM [24]	SkySense GPT
Scene Classification	ZS	SIRI-WHU	Acc	66.63	62.66	68.50	74.75
		AID	Acc	72.03	91.26	89.33	92.25
		WHU-RS19	Acc	86.47	93.17	97.00	97.02
		AID-multi	F1-Score	46.55	-	-	47.97
VQA	ZS	RSVQA-HR	Pre. Acc	58.45	-	65.00	69.14
			Comp. Acc	83.19	-	83.70	84.14
			Avg. Acc	70.82	-	74.35	76.64
	FT	RSVQA-LR	Rural. Acc	91.09	89.07	88.00	95.00
			Pre. Acc	90.33	88.51	89.58	91.07
			Comp. Acc	94.00	90.00	89.79	92.00
			Avg. Acc	91.81	89.19	89.12	92.69

下均实现了高准确率，表明其能够胜任复杂的细粒度任务，并具备对基础任务的强大理解能力。

6 结论

在本文中，我们构建了 FIT-RS，一个大规模的指令微调数据集，旨在提升 RSLMMs 的细粒度理解能力。它涵盖了各种任务和高质量语料库，包括基础和新颖的复杂任务。基于 FIT-RS，我们构建了 FIT-RSFG 基准。此外，我们构建了第一个基准 FIT-RSRC，用于评估 LMMs 在遥感关系理解方面的能力。通过将 FIT-RS 与其他多模态语料库结合，我们进一步介绍了 SkySenseGPT，它在不同粒度任务中表现出色，并在公开数据集中超越了当前的 RSLMMs。由于在具有挑战性的复杂任务中仍有改进空间，我们希望 FIT-RS 能有助于构建更强大的 RSLMMs。

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附录

A 概述

FIT-RS 数据集、数据生成脚本以及 SkySenseGPT 的模型权重将在`https://github.com/Luo-Z13/SkySenseGPT`开源，数据仓库的 DOI 为10.57967/hf/2529。在本附录中，我们提供了用于创建 FIT-RS 数据集中各种指令调优任务的模板和提示的详细描述。我们还提供了关于我们构建的数据集以及用于训练和测试的公共数据集的具体细节。此外，我们展示了详细的实验结果和全面的可视化结果。

B FIT-RS 数据集

我们详细解释了 FIT-RS 数据集中针对不同任务所使用的特定问题模板和提示，其具体格式如表 A1所示。在数据预处理过程中，由于采用传统的裁剪策略将原始大尺寸遥感图像（RSIs）分割为小图块，我们为详细图像描述任务设置了完整性阈值，即前景交并比（iof）阈值为 0.3，以提供全面的描述；而对于其他任务，则将该阈值设为 0.55，确保目标相关语料库的可靠性。

表 A1: FIT-RS 中的指令跟随任务格式。缩写及其全称如下：IC：图像描述，RC：区域描述，VQA：视觉问答系统，SC：场景分类，CC：复杂理解，MT：多轮对话。

Task Type	Task	Specific Format
IC	Detailed Image Caption	-
RC	Detailed Region Caption	-
VQA	Visual Question Answering	Answer in one word or a short phrase.
SC	Multi-Label Scene Classification	Classify the given image in one of the following classes. / Answer in one word or a short phrase. Classify the given image into the following classes. / Answer with all applicable classes separated by commas.
CC	Relation Detection	[detection]
	Relation Reasoning	[reasoning]
	Object Detection	[detection]
	Object Reasoning	[reasoning]
	Region-Level SGG	[grounding]
	Image-Level SGG	[grounding]
MT	Multi-Turn/Task Conversation	-

B.1 详细图片说明

前景标注生成。对于每张图像，场景图生成（SGG）的标注，即对象标注和三元组，通过自动化脚本转换为句子，作为 GPT-4/GPT-3.5 [1] 提示的一部分。目标检测标注以类似于 LLaVA [3] 的格式排列，不同之处在于我们采用了旋转边界框。对于三元组标注，我们不保留对象的 ID 号，因为它们是无序的，并且不可避免地会干扰 GPT 生成的最终描述。此外，

为了控制最终提示的长度，避免超过 GPT-4/GPT-3.5 的 token 限制，相同的三元组被合并为一个。例如，‘crane over ship, crane over ship, crane over ship’ 被转换为 ‘3 cranes over ship’，从而避免后续生成不完整的场景描述。

背景描述生成。由于原始场景图标注中缺乏对自然地物和建筑物的标注，我们利用现有的 LMMs 生成简洁而精确的背景描述。具体而言，使用强大且高效的 TinyLLaVA-3.1B [42]，我们能够以每张图像约 1.1 秒的速度快速生成大量 RSIs 的简洁背景描述。尽管 TinyLLaVA 在直接描述 RSIs 中的人造物体时面临幻觉问题，但它能准确描述水体、植被等主要背景元素。用于 TinyLLaVA 的提示词如表 A2的上半部分所示。在为 85,000 张分辨率为 512× 512 像素的 RSIs 生成背景描述后，我们手动随机审查了 2,000 张图像，确保了背景描述的基本准确率。

详细字幕生成。我们定义了一个系统提示，并将背景与前景描述结合作为用户提示。随后，它们被整合输入到 GPT-4/GPT-3.5 API 中，以获取流畅、详尽且全面的图像描述。所使用的提示及信息格式如表 A2所示。生成背景描述及最终详细图像字幕的提示见表 A3。

此外，我们在表 A4 和表 A5 中展示了示例，说明如何通过结合丰富详细的前景信息和简洁精确的背景信息，利用 GPT API 生成对整幅图像的全面描述。我们可以观察到，对象之间复杂的关系在解释对象状态和深入理解遥感场景中起着至关重要的作用。

表 A2: 对于每个查询，我们展示了为 GPT3.5/GPT-4 构建提示的过程，以从查询 [`上下文']中收集查询 [`响应']，采用 few-shot 上下文学习，其中示例来自**fewshot_ 样本**，每个示例包括输入样本 [`上下文']和输出样本 [`响应']。请注意，消息是最终的提示。查询 [`上下文']由三部分组成：**背景描述**、**对象**和**关系**，详情请参见表 A4和表 A5中的示例。

```
messages=[ {"role":"system", "content": f""" 你是一个能够分析单张图像的 AI 视觉助手。你会收到一个描述你所观察图像的背景字幕。此外，图像中特定物体的位置和关系也会被提供，同时附有详细的坐标。这些坐标以旋转边界框的形式给出，表示为 (x1, y1, x2, y2, x3, y3, x4, y4)，这些值对应于 rbox 的 4 个角点。

利用提供的字幕、边界框和关系信息，用自然语言详细描述场景。不要直接提及边界框坐标，而是利用这些数据来解释场景。包括物体数量、物体位置以及物体之间的关系等细节。

在使用字幕和坐标信息时，直接解释场景，不要提及信息来源是字幕或边界框，避免提到坐标的确切值和 ‘iof’。避免主观的、带有情感色彩的推测和描述。注意关系可能是相互的，因此不要根据关系来确定目标数量。回答时要像直接观察图像一样。不要让你的回答过长。"""}
] for _sample in fewshot_samples:
    messages.append({"role":"user", "content":sample[`context']})
    messages.append({"role":"assistant", "content":sample[`response']}
    )
messages.append({"role":" 用户", "content":`\n'.join(查询)})
```

表 A3: 详细图像说明的指令列表。

- "Describe this image in detail."
- "Provide a detailed description of the given image."
- "Analyze the image in a comprehensive and detailed manner."
- "Give a thorough description of the image."
- "Provide an in-depth analysis of the image."
- "Describe the elements present in this image in detail."
- "Give a detailed interpretation of the image."
- "Analyze and describe the image in detail."
- "Provide a comprehensive description of what you see in the image."
- "Describe the image thoroughly, including all the elements you can identify."
- "Give an extensive description of the image."
- "Provide a detailed explanation of the image's content."
- "Describe in detail what you observe in the image."
- "Give a comprehensive analysis of the image."
- "Elaborate on the details you can observe in this image."
- "Provide an in-depth description of every element in the image."
- "Detail the contents of the image."
- "Analyze and describe every detail you can identify in the image."
- "Give a detailed account of the image's content."

B.2 详细区域说明

在详细区域标注任务中，我们首先为目标创建一个基本描述，涵盖基本位置、基本大小、可见性及位置表示。随后，描述与之相关的所有三元组。

基础定位。图像最初被划分为 9 个基本区域。然后，根据目标中心点（旋转边界框的中心）在这些区域中的位置来确定目标的位置。可能的位置包括“中心”、“左上”、“右上”、“左下”、“右下”、“上”、“下”、“左”和“右”。

基本大小。根据遥感图像中小目标检测的基本定义 [66]，目标基于 16 × 16、64 × 64 和 256 × 256 像素的阈值被分类为“小”、“中”、“大”和“巨型”。

基本可见性。根据裁剪后目标的交并比（iof）值，我们将其分类为：当 iof 值小于 0.4 时为“勉强可见”，0.4 到 0.6 之间为“部分可见”，0.6 到 1.0 之间为“清晰可见”，而 iof 值等于 1.0 时为“完全可见”。需要注意的是，在创建区域相关任务时，旋转的边界框可能会包含部分背景。为了减少噪声的引入，我们设置了一个相对较高的 iof 阈值（0.55）。因此，参与后续复杂理解任务的目标大多为部分可见。

表 A4: 一个例子展示了背景与前景描述的融合，以及最终生成的详细描述。

Background Caption:
The airport runway is surrounded by grass.

Objects:
apron: [10.16, 67.77, 40.62, 97.85, 29.49, 110.94, -1.76, 80.66], iof: 0.85
taxiway: [73.63, 31.84, 50.0, 56.25, 57.81, 63.67, 81.64, 38.87], iof: 1.0
airplane: [88.09, 46.29, 91.99, 42.19, 96.88, 47.66, 92.19, 51.56], iof: 1.0
airplane: [93.75, 13.67, 98.63, 18.55, 93.75, 22.66, 89.26, 18.36], iof: 1.0
airplane: [11.91, 86.91, 8.59, 83.2, 12.89, 78.32, 16.6, 81.84], iof: 1.0
airplane: [7.23, 81.64, 3.91, 78.52, 7.03, 75.0, 10.55, 77.93], iof: 1.0
taxiway: [44.73, 58.4, 34.57, 117.77, 40.23, 117.77, 49.02, 59.38], iof: 0.67
Relationships:
1 taxiway through apron, 2 airplanes parallelly parked on apron, 1 airplane parking in the same apron with airplane, 1 airplane parked alongside with airplane.

Final Image Caption:
In the image, we observe an airport scene featuring a runway surrounded by lush grass, illustrating a typical serene airport environment. The scene includes various key components such as an apron, taxiways, and airplanes, arranged in a coordinated manner. The apron, which serves as a parking area for airplanes, is prominently positioned and appears to have ample space. It is somewhat rectangular but with an irregular shape, indicating how it accommodates the specific layout and needs of the airport. There are two taxiways visible in the scene. One taxiway is located away from the apron, well-defined and appears to be facilitating the movement of aircraft between the runway and the apron. This taxiway has a clear, well-maintained pathway, indicating regular usage. The other taxiway is closer to the apron with a more complex shape, possibly due to the terrain or specific design requirements of the airport. Despite its complexity, it plays a crucial role in aircraft navigation on the ground. Four airplanes are spotted within the scene, each at different positions but primarily associated with the apron area. Two of these airplanes are parallelly parked on the apron, showcasing an organized and efficient use of space. These airplanes are aligned in such a way that maximizes the apron's capacity while ensuring safety and ease of movement for ground vehicles and personnel. The other two airplanes, although individually mentioned, seem to be parked in close proximity to each other on the same apron, hinting at a busy operational day at the airport with multiple aircraft being serviced or prepared for flights. The arrangement of these airplanes alongside each other suggests a well-coordinated effort to manage space and operational activities on the apron efficiently. Overall, the layout and operational dynamics of the airport components in the image – from the design of the taxiways facilitating movement to the organized parking of airplanes on the apron – display a well-managed and functional airport environment conducive to safe and efficient air travel operations.

表 A5: 一个示例，展示了结合背景和前景描述，以及生成的最终详细描述。

Background Caption:
The image shows a city street with a lot of trees and buildings.

Objects:
car: [57.23, 12.89, 54.88, 6.25, 51.56, 7.42, 53.91, 14.06], iof: 1.0
car: [95.7, 77.73, 98.05, 84.18, 101.56, 83.01, 99.22, 76.56], iof: 0.85
car: [83.59, 38.09, 85.74, 44.92, 88.67, 44.14, 86.72, 37.3], iof: 1.0
car: [15.23, 98.44, 13.87, 92.38, 11.13, 92.97, 12.5, 99.02], iof: 1.0
car: [12.11, 99.61, 9.57, 94.14, 6.84, 95.31, 9.38, 100.78], iof: 0.95
car: [6.84, 101.76, 5.86, 95.12, 2.93, 95.51, 3.91, 102.34], iof: 0.69
car: [10.55, 86.91, 7.23, 81.05, 4.49, 82.42, 7.81, 88.28], iof: 1.0
car: [9.57, 80.47, 11.13, 87.11, 14.06, 86.33, 12.5, 79.69], iof: 1.0
car: [6.25, 88.09, 3.91, 82.23, 1.17, 83.2, 3.32, 89.06], iof: 1.0
car: [24.02, 101.17, 27.73, 100.59, 27.34, 98.05, 23.63, 98.63], iof: 0.65
car: [22.85, 98.24, 25.98, 97.66, 25.59, 95.12, 22.27, 95.7], iof: 1.0
car: [49.22, 63.28, 47.27, 57.03, 44.34, 57.81, 46.29, 64.26], iof: 1.0
car: [44.73, 48.44, 42.58, 41.6, 39.84, 42.58, 41.99, 49.22], iof: 1.0
truck: [58.01, 65.43, 52.73, 51.37, 48.83, 52.93, 54.3, 66.99], iof: 1.0
truck: [49.02, 39.06, 44.34, 24.61, 40.43, 25.98, 45.12, 40.23], iof: 1.0

Relationships:
3 cars driving in the same direction with car, 3 cars driving in the same lane with car, 3 cars within safe distance of car, 7 cars parked alongside with car, 2 trucks driving in the same direction with truck, 2 trucks driving in the same lane with truck, 2 trucks within safe distance of truck.

Final Image Caption:
The scene depicts a bustling city street graced with an abundance of trees and buildings. Along this street, numerous vehicles are present, showcasing a typical vibrant urban environment. Specifically, there is a variety of cars in different states of motion and rest. Some cars are on the move; a pair of these are driving in the same direction and sharing the same lane, maintaining a safe distance from one another. This indicates a flow of traffic on the street. Additionally, there are cars that appear to be parked on the side of this street. A sequence of these vehicles is aligned, suggesting a designated parking area. These parked cars are situated alongside each other, adhering to what seems to be an orderly arrangement according to parking protocols. Moreover, two trucks are part of the street's vehicle ecosystem. They are larger than the other vehicles present and are also in motion, following the traffic pattern. These trucks are driving in the same direction, staying within the same lane, and ensuring there is a safe distance between them, which points to the drivers' adherence to road safety measures. The visibility of each vehicle in the street scene varies, with some cars fully visible, indicating clear, unobstructed views, while others have varying degrees of partial visibility, which could be due to their placement behind other objects in the environment or the viewing angle. This typical urban snapshot represents the organized chaos of city traffic, where various vehicles share the road, adhere to driving regulations, and coexist within the street's vibrant landscape.

在获取上述基本描述后，可以将其组合起来描述用户指定的区域，即区域标注任务的输入。组合后的描述能够翻译目标的基本类别、大小、位置和可见性，例如“图像右下角完全可见的中型起重机”。接下来，我们在场景图标注中搜索与目标相关的所有三元组，并将其转换为自然语言格式，将目标称为“它”，以形成连贯且全面的区域描述。问题指令列表可在表 A6 中找到。

B.3 视觉问答系统

我们精心设计了全面且多样化的问题模板用于 VQA，如表 A7所示。对于每种类型的问题，我们从模板中随机选取一个问题，然后根据图像的场景图标注随机选择相应的类别或关系。这些内容填入模板中的“<>”，生成最终的问题并创建相应的答案。下面，我们概述了不同类型问题的构建方法。

存在性问题。我们生成问题以确定图像中每个对象类别的存在性。对于存在的类别，我们创建“是”问题来确认其存在。对于不存在的类别，我们生成“否”问题以表明其缺失。我们的目标是平衡生成的“是”与“否”问题的数量，确保每个类别在提问过程中得到充分体现。

比较问题。我们生成问题以比较图像中不同物体类别的数量。对于每一对存在的类别，我们创建问题，询问一个类别的实例数量是否多于、少于或等于另一个类别。这些问题旨在评估图像中识别类别的相对数量，从而提供关于它们比较存在情况的洞察。每个问题都附有一个基于图像中类别实际计数的答案（“是”或“否”）。目标是提供场景中不同物体类别相对数量的平衡评估。

计数问题。我们生成问题以询问图像中特定实体类别和关系的数量。对于图像中检测到的每个对象类别，我们构建询问实例数量的问题。每个问题都与其对应的数字答案配对，反映图像中对象的实际数量。此外，我们从实体之间的关系中选择并创建关注特定关系的问题。这些问题询问由指定关系连接的主客体对的数量。每个问题都基于计算出的符合指定标准的主客体关系数量来回答。目的是提供对图像中单个对象类别数量及其之间关系的详细理解。

关系问题。我们生成问题以探究图像中检测到的对象类别之间关系的存在性和特异性。当识别出有效的关系三元组时，我们继续构建问题，重点关注这些关系的存在性和一致性。这些问题询问图像中特定对象类别的所有实例是否与另一个对象类别存在指定的关系。这些问题的答案取决于主体类别的所有实例是否始终通过指定关系与对象类别的实例相连。还可能生成额外的问题，以确认是否存在满足这些关系标准的实例。

此外，如果未检测到有效的关系三元组，我们会创建问题来验证随机选择的对象类别对之间是否存在特定关系。这些问题旨在确认图像中是否缺少某些类型的对象和关系。每个问题都以**否定**的方式回答，以反映此类关系的缺失。

B.4 复杂理解任务

除了主论文中提供的基本解释外，我们相信在此提出的复杂理解任务在多个方面超越了现有的区域级任务。用于构建关系检测、关系推理、目标检测和目标推理任务问题的指令模板如表 A8 和表 A9 所示。对于区域级 SGG 和图像级 SGG 任务，我们分别使用相同的指令进行详细区域描述和详细图像描述，但为它们添加了前缀 [grounding]。更多视觉示例可见图 A1。

表 A6: 详细区域标注说明列表。

- ”Can you provide me with a detailed description of the region in the image marked by <region>?”
- ”I’m curious about the region represented by <region> in the picture. Could you describe it in detail?”
- ”What can you tell me about the region indicated by <region> in the image?”
- ”I’d like to know more about the area in the photo labeled <region>. Can you give me a detailed description?”
- ”Could you describe the region shown as <region> in the picture in great detail?”
- ”What details can you give me about the region outlined by <region> in the photo?”
- ”Please provide me with a comprehensive description of the region marked with <region> in the image.”
- ”Can you give me a detailed account of the region labeled as <region> in the picture?”
- ”I’m interested in learning more about the region represented by <region> in the photo. Can you describe it in detail?”
- ”What is the region outlined by <region> in the picture like? Could you give me a detailed description?”
- ”Can you provide me with a detailed description of the region in the picture marked by <region>, please?”
- ”Please provide a detailed description of the region <region> in the given image.”
- ”What can you tell me about the region indicated by <region> in the image, exactly?”
- ”I’d like to know more about the area in the photo labeled <region>, please. Can you give me a detailed description?”
- ”Could you describe the region shown as <region> in the picture in great detail, please?”
- ”Please provide me with a comprehensive description of the region marked with <region> in the image, please.”
- ”Can you give me a detailed account of the region labeled as <region> in the picture, please?”
- ”I’m interested in learning more about the region represented by <region> in the photo. Can you describe it in detail, please?”
- ”What is the region outlined by <region> in the picture like, please? Could you give me a detailed description?”
- ”Please describe the region <region> in the image in detail.”
- ”Can you offer a thorough analysis of the region <region> in the image?”
- ”Could you elaborate on the region highlighted by <region> in the picture provided?”
- ”Please share more information about the zone emphasized with <region> in the photo.”

表 A7: 为生成视觉问答问题，模板会遍历各类图像标注，如（类别 1，类别 2）或（类别 1，关系）。针对每种问题类型，我们随机选取一个模板，并用当前遍历到的内容进行填充。

Presence Question:

Type 1 "Is there a/an <cat name> in the image?"
Type 2 "Does the image contain a/an <cat name>?"
Type 3 "Can a/an <cat name> be found in the image?"

Compare Question:

Type 1 "Are there fewer <cat1 name> than <cat2 name> in the image?"
Type 2 "Is the number of <cat1 name> greater than the number of <cat2 name>?"
Type 3 "Does the image contain the same number of <cat1 name> and <cat2 name>?"

Count Question:

Type 1 "How many <cat name>s are there in the image?"
Type 2 "What is the number of <cat name> in the image?"
Type 3 "What is the amount of <cat name> in the image?"
Type 4 "What is the count of <subj cat name>s in the image that are <rel name> a/an <obj cat name>?"
Type 5 "What is the amount of <subj cat name> in the image are <rel name> a/an <obj cat name>?"
Type 6 "How many <subj cat name>s in the image are <rel name> a/an <obj cat name>?"

Relation Question:

Type 1 "Are all <subj cat name>s in the image <rel name> a/an <obj cat name>?"
Type 2 "Does the image show all <subj cat name>s <rel name> a/an <obj cat name>?"
Type 3 "Is every <subj cat name> in the image <rel name> a/an <obj cat name>?"

表 A8: 用于构建关系检测与关系推理任务的模板，以及对应的否定答案。模板中特殊符号内的内容将被替换为具体的目标信息或关系。

Templates of Relation Detection Task:

- What relationships exist between <category1> and <category2> in the image?
- Describe all existing relationships between <category1> and <category2> in the image.
- Describe all relationships between <category1> and <category2> in the image.
- Detect all existing relationships between <category1> and <category2>.
- Could you identify and explain the relationships between <category1> and <category2> in the image?
- Please analyze and describe all the relationships between <category1> and <category2> depicted in the image.
- Give all relationships existing between <category1> and <category2> in the image.
- Could you please describe all the relationships between <category1> and <category2>?
- In this image, can you describe all the relationships between <category1> and <category2> for me?

Templates of Relation Reasoning Task:

- Please identify the relationship between <subject in> <region1> and <object in> <region2> in the image, and output their categories.
- Describe the relationship between <subject in> <region1> and <object in> <region2> in the image, and output their categories.
- What is the relationship between <subject in> <region1> and <object in> <region2> in the image? And output their categories.
- Could you identify the relationship between <subject in> <region1> and <object in> <region2> in the image? And output their categories.
- What type of relationship exists between <subject in> <region1> and <object in> <region2> in the image? And output their categories.
- What kind of relationship is illustrated between <subject in> <region1> and <object in> <region2> in the image? And output their categories.
- Could you help me understand the relationship between <subject in> <region1> and <object in> <region2> in the image? And output their categories.

Templates of Negative Answer:

Type 1 "I'm sorry, I cannot answer as the given image does not contain any given objects."
Type 2 "I'm sorry, I cannot answer as the relationship is unclear in the image."
Type 3 "I'm sorry, I cannot answer as there are no objects that satisfy the conditions."

表 A9: 用于构建目标检测与目标推理任务语料库的模板。模板中特殊符号内的内容将被替换为具体的目标信息或关系。

Templates of Object Detection Task:

- Can you locate all the <category> in the image?
- Could you help me find all the <category> in the image? Please provide their locations.
- Detect all the <category> in the image and output their locations.
- Detect all the <category> and output their locations.
- Provide the coordinates of all <category> in the image.
- Can you find and mark the positions of all the <category> in the given image?
- Please detect all the <category> in the image and output their locations.
- Locate and list the positions of all <category> that appear in the image.
- Identify and provide the coordinates of all <category> in the image.
- Identify all the <category> and mark their locations.
- I need you to detect and locate all <category> present in the image.
- Detect the locations of all <category> objects in the provided image.
- Please locate all the <category> in the given image.

Templates of Object Reasoning Task:

- Assist me in locating and classifying all the <subject> <relation> the <object>[in <region>].
- I want to know the coordinates and categories of all the <subject> <relation> the <object>[in <region>].
- There are some <subject> that are <relation> the <object>[present at <region>]. Could you tell me their locations and categories?
- Please locate and categorize all the <subject> that are <relation> the <object>[in location <region>].
- Find all the <subject> that have a relationship of <relation> with the <object>[present at <region>]. Can you give me their positions and categories?
- Your task is to locate all <subject> that have a relationship <relation> with <object>[in location <region>] and classify them.
- I need you to locate and categorize all <subject> that have a relationship <relation> with <object>[in location <region>].
- Could you help me find all the <subject> that have a relationship of <relation> with the <object>[in <region>]? Please provide their locations and categories.
- Provide the coordinates and categories of all <subject> that are <relation> the <object>[present at <region>] in the image.
- Find the <subject> that have a relationship of <relation> with the <object>[in <region>]. Can you give me their positions and categories?
- What <subject> have the relationship of <relation> with the <object>[present at <region>]? Could you locate and classify them for me?
- Please find and classify all <subject> that has a relationship <relation> with

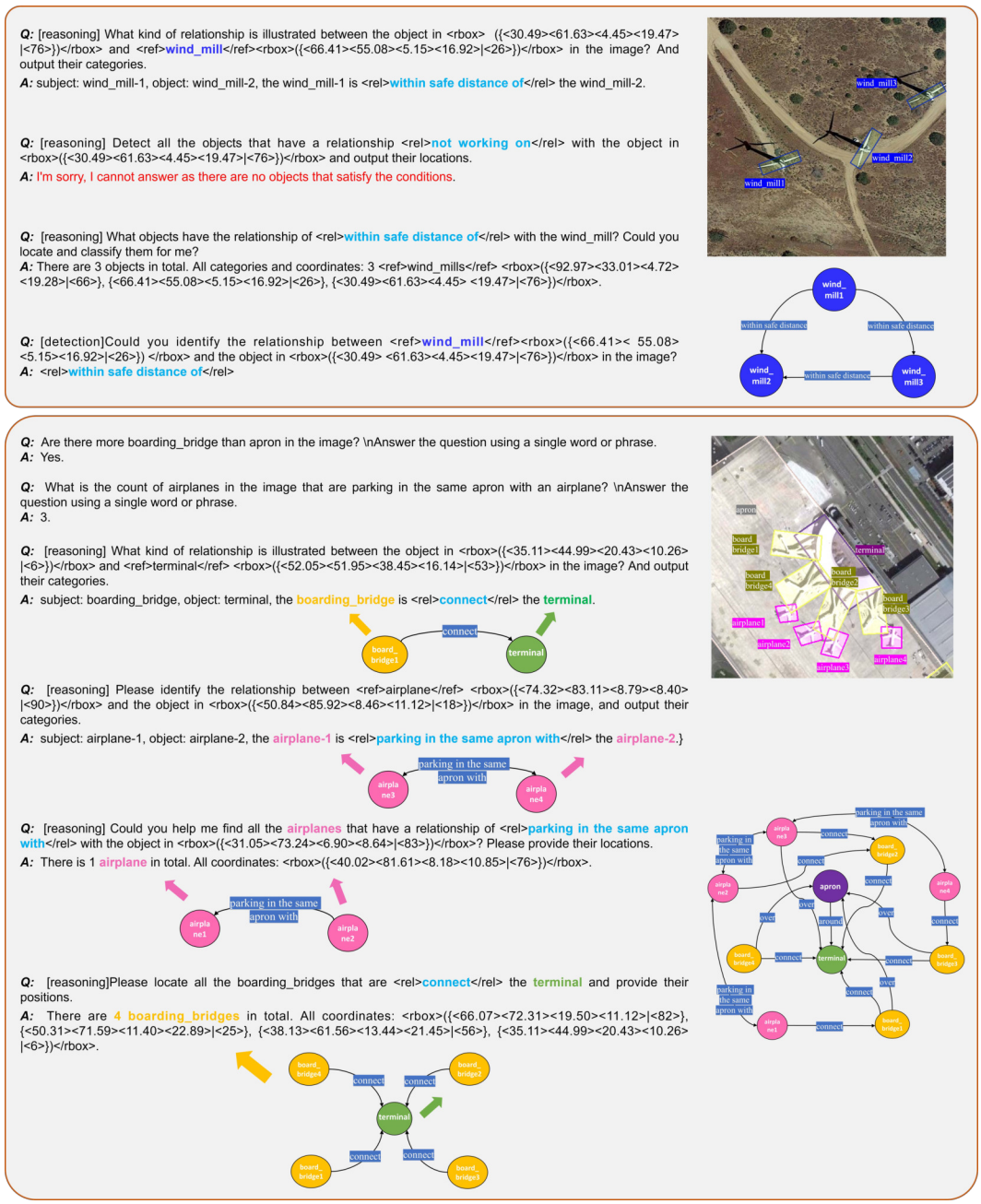

图 A1: 一个针对单张图像的多轮对话示例。可见，通过利用用户输入的 Region 和 Relation 作为辅助，Model 能够基于感知的场景图识别出 Target。

C 关于数据集的详细信息

C.1 额外训练数据集

此外，我们从现有的公开数据集中收集并转换了约 365k 条指令样本，具体来源数据量如表 A10所示，具体的转换方法如下所述。对于 GeoChat-Instruction [25]，我们基于 DOTA [41]、DIOR [55] 和 FAIR1M [56] 数据集，选取了其图像描述和多轮对话部分，以增强对图像的整体感知能力。

表 A10: 附加数据集中指令调优样本的详细信息。缩写及其全称如下：SC：场景分类，VQA：视觉问答系统，OD：目标检测，IC：图像描述，MT：多轮对话。

Dataset	Samples	Task Type	Specific Format
NWPU-RESISC-45 [50]	9000	SC	Classify the given image in one of the following classes.
UCM-Landuse [51]	2100		
RSITMD [52]	948		
EarthVQA [53]	145368	VQA	Answer in one word or a short phrase.
RSVQA-LR [43]	47173		
FloodNet-VQA[54]	4056		
DOTA-v2.0 [41]	20000	OD	[detection]
FAIR1M [56]	40000		
Geochat-Instruction [25]	96355	IC, MT	-

C.2 评估数据集

用于评估的公开数据集详情如表 A11 所示。若数据集具备明确定义的测试集，我们则采用其测试集进行 zero-shot 或微调性能评估。否则，通常使用整个数据集进行评估。

表 A11: 用于评估的公共数据集详情。

Dataset	Task Type	Image Size	Resolution (m)	Categories
SIRI-WHU [62]	SC	200×200	2	12
AID [63]		600×600	0.5-8	30
WHU-RS19 [64]		600×600	0-0.5	19
AID-multi [65]		600×600	0.5-8	17
RSVQA-HR [43]	VQA	256×256	10	-
RSVQA-LR [43]		512×512	0.15	-

D 基准的详细结果

首先，我们提供实验设置的更多细节。我们使用 LoRA 微调，秩 r 设为 64。训练时使用了 4 个 GPU，总批量大小设为 128，学习率为 1×10^{-6}。采用了 'cosine' 学习率调度，预热比例为 0.03，训练进行了 1 个轮次。

D.1 FIT-RSFG 基准

我们列出了基于 FIT-RS 数据集和所用评估指标的 FIT-RSFG 基准的详细结果。详细结果如表 A12所示。考虑到在 SGG 任务中，三元组匹配更强调类别的正确性，用于评估 mAP 指标的 IoU 阈值设定为 0.25。经过微调后，模型在一些具有挑战性的任务上表现显著提升。此外，整体准确率分布与我们为任务设计的渐进难度相符。

D.2 FIT-RSRC 基准

在 FIT-RSRC 基准中，我们构建了四种类型的问题（关系、对象、主语、存在），每种类型各占 25%，每种类型内的答案都有负答案分布。每个问题选项的长度为 4-5，其中 87.5% 的问题有 4 个选项，12.5% 的问题有 5 个选项。为了更好地分析评估的 LMMs 的预测准确率，我们统计了所有 4 选项问题的答案分布，如图 A2所示。可以看出，现有的 LMMs 表现出对某个选项（选项 D）的偏见，而 SkySenseGPT 表现出较少的偏见。

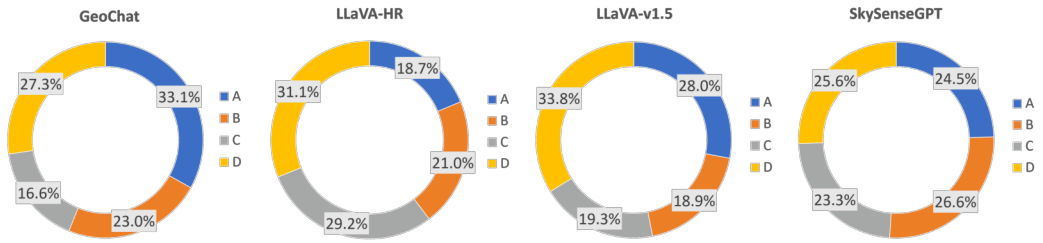

图 A2: 不同 LMMs 预测的选择分布，均采用 CirularEval 记录。

此外，我们展示了由人类专家在 GPT-4 协助下精心制作的预设常识词表。这些词表包括主语-关系和关系-宾语词表，如表 A13和表 A14所示。在生成高质量干扰选项时，所有生成的内容都会与这些词表进行交叉核对，以确保错误选项不能仅凭文本内容被识别。这保证了 FIT-RSRC 基准的高质量。

表 A12: FIT-RSFG 基准的详细结果。对于详细的图像字幕，测试集选自 GPT-4 生成的描述。Count-Acc、Pre-Acc、Comp-Acc 和 Rel-Acc 分别代表四类问题（计数、呈现、比较和关系）的准确率。

Task Type	Task	Metric	LLaVA1.5-7B (ZS)	GeoChat (ZS)	SkySenseGPT (FT)
IC	Detailed Image Caption	BLEU-1	15.38	8.79	27.31
		BLEU-2	13.05	4.36	15.64
		BLEU-3	6.23	1.88	7.95
		BLEU-4	2.93	0.83	4.32
		METEOR	10.55	4.23	12.91
		ROUGE_L	20.57	10.55	23.16
		GPT-eval	4.30	4.06	6.00
RC	Detailed Region Caption	BLEU-1	14.70	14.55	75.82
		BLEU-2	6.63	8.66	71.42
		BLEU-3	2.33	4.63	67.71
		BLEU-4	0.63	1.97	64.05
		METEOR	10.66	8.80	43.69
		ROUGE_L	15.67	14.02	71.86
VQA	Visual Question Answering	Count-Acc	27.75	31.64	46.49
		Pre-Acc	81.91	51.70	92.82
		Comp-Acc	55.29	64.13	82.54
		Rel-Acc	69.40	66.40	90.18
		Ave Acc	58.59	53.47	79.76
SC	Multi-Label Scene Classification	Scene Acc	56.18	42.27	82.23
		Obj-Precision	19.04	9.10	77.28
		Obj-Recall	9.69	41.89	66.40
		Obj-F1-Score	12.84	14.95	71.43
CC	Relation Detection	F1-Score	Failed	Failed	88.68
	Relation Reasoning	F1-Score	Failed	Failed	74.33
	Object Detection	mAP	Failed	6.13	27.40
	Object Reasoning	mAP	Failed	Failed	5.71
	Region-Level SGG	Recall	Failed	Failed	17.01
		Mean Recall	Failed	Failed	4.05
	Image-Level SGG	Recall	Failed	Failed	9.60
		Mean Recall	Failed	Failed	3.59
		mAP	Failed	Failed	12.99
MT	Multi-Turn Conv	GPT-eval	2.74	3.64	6.60

表 A13: 专家借助 GPT-4 创建的主语-关系词表。FIT-RSRC 中的所有干扰项均需存在于该列表中，以确保 LMM 无法仅凭文本推导出答案。

Commonsense Subject-Relation List:

storehouse: adjacent, connect, over, co-storage with, around, not co-storage with
breakwater: connect, over, adjacent, through, converge, intersect, around
dock: connect, over, adjacent, through, converge, intersect, around
airplane: connect, over, within safe distance of, parallelly parked on, isolatedly parked on, randomly parked on, run along, not run along, parking in the same apron with, parking in the different apron with, parked alongside with, not parked alongside with, running along the different taxiway with, running along the same taxiway with, approach, away from, within danger distance of, pass under, pass through, pass across
boarding bridge: connect, over, adjacent, through, converge, intersect, around
runway: connect, over, adjacent, through, converge, intersect, around
taxiway: connect, over, adjacent, through, converge, intersect, around
terminal: connect, over, adjacent, through, converge, intersect, around
apron: connect, adjacent, through, converge, intersect, around
truck: within safe distance of, driving in the same direction with, parallelly parked on, isolatedly parked on, parked alongside with, not parked alongside with, approach, away from, within danger distance of, incorrectly parked on, in the same parking with, in the different parking with, driving in the opposite direction with, driving in the different lane with, driving in the same lane with, driving alongside with, drive toward, pass across, drive off
car: within safe distance of, driving in the same direction with, parallelly parked on, isolatedly parked on, parked alongside with, not parked alongside with, approach, away from, within danger distance of, incorrectly parked on, in the same parking with, in the different parking with, driving in the opposite direction with, driving in the different lane with, driving in the same lane with, driving alongside with, drive toward, pass across, drive off
cooling tower: violently emit, slightly emit, exhaust to, supply to
chimney: violently emit, slightly emit
vapor: connect
smoke: connect
genset: over, exhaust to, supply to, directly transmit electricity to
coal yard: adjacent, supply to, over, through, intersect
lattice tower: within different line of, within same line of, directly connected to, indirectly connected to, directly transmit electricity to, indirectly transmit electricity to
substation: within same line of, directly connected to, directly transmit electricity to, indirectly transmit electricity to
wind mill: within safe distance of
flood dam: adjacent, over, connect, intersect
ground track field: around, over, through, converge, intersect
basketball court: around, over, through, converge, intersect
engineering vehicle: working on, not working on
soccer ball field: around, over, through, converge, intersect
tennis court: around, over, through, converge, intersect
tower crane: working on, not working on
containment vessel: co-storage with, not co-storage with, supply to

表 A14: 由专家在 GPT-4 协助下创建的部分关系-对象词汇表。FIT-RSRC 中的所有干扰选项均需存在于该列表中，以确保 LMM 无法仅凭文本推断出答案。

Commonsense Relation-Object List:

parallelly docked at: dock, breakwater
isolatedly docked at: dock, breakwater
connect: tank, storehouse, breakwater, dock, boarding bridge, runway, taxiway, terminal, apron, vapor, smoke, substation, toll gate, bridge, intersection, roundabout, genset, cooling tower, chimney, lattice tower, wind mill, containment vessel
over: ship, boat, crane, goods yard, breakwater, dock, runway, taxiway, terminal, apron, truck parking, car parking, bridge, genset, coal yard, cement concrete pavement, flood dam, gravity dam, ground track field, basketball court, foundation pit, intersection, soccer ball field, tennis court, arch dam, roundabout, baseball diamond, stadium
co-storage with: containment vessel, tank, storehouse, goods yard, coal yard, substation, genset, cooling tower, chimney
within safe distance of: ship, boat, airplane, truck, car, wind mill
randomly docked at: dock, breakwater
docking at the same dock with: ship, boat
docked alongside with: ship, boat
docking at the different dock with: ship, boat
driving in the same direction with: ship, boat, engineering vehicle, truck, car
parallelly parked on: runway, taxiway, apron, truck parking, car parking
isolatedly parked on: runway, taxiway, apron, truck parking, car parking
randomly parked on: runway, taxiway, apron, truck parking, car parking
run along: runway, taxiway
adjacent: flood dam, crane, goods yard, storehouse, breakwater, dock, runway, taxiway, terminal, apron, genset, coal yard
through: runway, taxiway converge: runway, taxiway intersect: runway, taxiway
not run along: runway, taxiway
parking in the same apron with: airplane
parking in the different apron with: airplane
parked alongside with: engineering vehicle, crane, airplane, truck, car
not parked alongside with: engineering vehicle, crane, truck, airplane, car
running along the different taxiway with: airplane, engineering vehicle, crane, truck, car
around: ground track field, basketball court, tank, soccer ball field, taxiway, terminal, car parking, truck parking
not co-storage with: containment vessel, tank, storehouse
running along the same taxiway with: airplane, engineering vehicle, crane, truck, car
approach: intersection, dock, breakwater, runway, taxiway, roundabout, apron, gas station, bridge, cement concrete pavement, toll gate
away from: ship lock, intersection, dock, breakwater, runway, taxiway, roundabout, apron, gas station, truck parking, car parking, bridge, cement concrete pavement, toll gate
within danger distance of: ship, boat, airplane, truck, car
running along the different runway with: airplane, engineering vehicle, crane, truck, car

incorrectly parked on: truck parking, car parking
in the same parking with: truck, car, engineering vehicle, crane, airplane
in the different parking with: truck, car, engineering vehicle, crane, airplane